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SUPERSEDING
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MILITARY SPECIFICATION

VINYL ALCOHOL – ACETATE RESIN SOLUTION (VAAR) (FOR USE IN AMMUNITION)

Reactivated after 07 October 2020 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the U.S. Army Combat Capabilities Development Command – Armaments Center (CCDC – AC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This Military Specification covers one type of Vinyl Alcohol Acetate Resin Solution for use as a binder in pyrotechnic compositions (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

DEPARTMENT OF DEFENSE STANDARDS

DELETED

(Copies of these documents are available online at https://quicksearch.dla.mil/)

Comments, suggestions, or questions on this document should be addressed to the U.S. Army Combat Capabilities Development Command Armaments Center (CCDC AC), ATTN: FCDD-ACE-QSA, Picatinny Arsenal, NJ 07806 or emailed to usarmy.picatinny.list.ccdc-ac-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC: 1376

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids should apply.

CODE OF FEDERAL REGULATIONS

49 CFR - 171-190 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission relations are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, Government Printing Officer, Washington, D.C. Orders for the above publications should cite: "49 CFR 171-190 (latest revision)."

3. REQUIREMENTS

3.1 Material. The vinyl alcohol acetate resin solution is a mixture of approximately 28 percent solids and 72 percent solvent. The solids consist of approximately 18 percent hydroxyl content by weight, calculated as vinyl alcohol, and 82 percent vinyl acetate. The solvent is technical grade methyl acetate, 82 percent purity.

3.1.1 Color. The color of the vinyl alcohol acetate resin solution shall be equivalent to a No. 300 platinum cobalt standard when tested as specified in 4.3.1.

3.1.2 Polyvinyl alcohol. The polyvinyl alcohol content shall be 16 to 19 percent by weight calculated on a dry resin basis when tested as specified in 4.3.2.

3.1.3 Total solids. The total solid content shall be 26 to 29 percent by weight when tested as specified in 4.3.3.

3.2 Workmanship. The vinyl alcohol acetate resin solution shall be free from impurities, such as wood chips, water, oil, and other solvents not specified in 3.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 DELETED

4.1.1 Submission of product. At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product submitted:

(a) A statement that the lot complies with all of the quality assurance provisions specified within this specification.

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(b) Drawing, specification number and date, together with an identification and date of changes thereto.

(c) A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

(d) Number of items in the lot.

(e) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions.

4.2.1 Lot formation. The term "lot", as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot.

4.2.2 Testing.

4.2.2.1 Sample. A random sample of 10 containers shall be selected from each lot, all containers being of the same size. When lots are comprised of 10 containers or less, each container shall be sampled.

4.2.2.2 Preparation of composite samples. Four-ounce samples of VAAR shall be removed from each of the ten containers selected to forma composite sample. The four ounces selected shall be blended to form two 20-ounce composite samples. The two samples shall be tested in accordance with the requirements of 3.1 through 3.2. If either of the composite samples fails to meet the requirements specified, the lot shall be rejected.

4.3 Test methods and procedures.

4.3.1 Color, Major Defect, Code No. 01001 (see 6.2).

4.3.1.1 Preparation of the No. 500 Platinum-Cobalt Standard. Measure 500 ml of distilled water into a 1000-ml volumetric flask. Add 100 ml concentrated hydrochloric acid and

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mix well. Weigh 1.245 gm of potassium chloroplatinate, K_2PtCl_6, and 1.000 gm of cobaltous chloride, $CoCl_2 \cdot 6H_2O$, to the nearest mg, and transfer to the flask. Swirl until complete solution is effected. Dilute the solution in the flask to the mark with distilled water and mix thoroughly. The color of this standard solution is equivalent to 500 platinum-cobalt units (500 mg of metallic platinum per liter), or each ml of this standard contains 0.5 mg of metallic platinum.

4.3.1.2 Preparation of the Platinum-Cobalt color standards. Prepare the required color standards by diluting the No. 500 platinum-cobalt standard solution as shown in the following table. Usually an experienced analyst can make a sufficient estimation of the color from these standards. If more exact color comparison is desired, prepare additional standards to supplement those given below. One color unit is equivalent to one mg of metallic platinum per liter.

TABLE I.

Platinum-Cobalt Color Standard Number	Platinum-Cobalt Standard, ml	Distilled Water, ml
200	40.0	60.0
300	60.0	40.0
400	80.0	20.0

4.3.1.3 Procedure. Transfer 100 ml of the sample to one of two matched 100-ml tall-form Nessler tubes. Fill the second tube to the mark with platinum-cobalt standard which seems to match the color of the sample, as indicated by a preliminary estimation. Compare the colors to the sample and the standard by viewing vertically down through tubes against a white background. Replace the liquid in the second tube with lighter or darker standards until an exact match is obtained.

4.3.2 Polyvinyl alcohol, Major Defect, Code No. 02001. Prepare and store in 1-pint screw-cap bottles the series (or such part of the series needed) of water-acetone mixtures shown in the table below. To 30 ml of one of the mixtures in a test tube, add 4 drops of the sample resin and shake. Continue until two successive mixtures are found, in one of which the resin forms a clear solution and in the other of which the resin forms a cloud or haze. Read from the table of the weight percent polyvinyl alcohol, on the dry resin basis, corresponding to the mixture in which the resin forms a cloud.

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TABLE II.

TABLE OF MIXTURES		
Percent by Weight Polyvinyl Alcohol; Dry Resin Basis	Mixture	
	Water, ml.	Acetone, ml.
15.0	206	194
15.5	208	192
16.0	210	190
16.5	214	186
17.0	216	184
17.5	220	180
18.0	222	178
18.5	226	174
19.0	228	172
19.5	232	168
20.0	234	166
20.5	238	162

4.3.3 Total solids, Major Defect, Code No. 03001. Weight rapidly into a tared, three-inch, aluminum dish, 3 to 5 gm of the sample resin to the nearest mg. Add 20 ml of SD-2B alcohol to the dish, place in a steam oven maintained at 135 +/- 1°C, and allow the SD-2B alcohol to evaporate. After the evaporation of SD-2B alcohol is complete, allow the dish to remain in the oven for one hour. Cool to room temperature and obtain the dry weight of the sample resin. Calculate as follows:

$$\frac{gm\ dried\ sample}{gm\ original\ sample} * 100 = total\ solids, \%\ by\ weight$$

5. PREPARATION FOR DELIVERY

5.1 Packing. Vinyl alcohol acetate resin shall be packed and shipped in such a manner as to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery. Containers shall conform to the ICC Regulation.

5.2 Marking. Marketing to insure safe handling shall conform to the ICC Regulations stipulated in the Code of Federal Regulations.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.2 Inspection code numbers. The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

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6.3 Test results. A copy of the test results should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND2, Dover, New Jersey, 07801

6.4 VAAR. An approved source of supply for VAAR is Union Carbide Plastic Company, 270 Park Avenue, New York City, New York, 10017. The VAAR has a designation number of MA-28-18.

6.5 Key words. Binder
Composition
Pyrotechnic

6.6 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodian:
Army – AR

Preparing Activity:
Army – AR

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